

LAPORAN
UAS PRAKTIKUM BASIS DATA 2

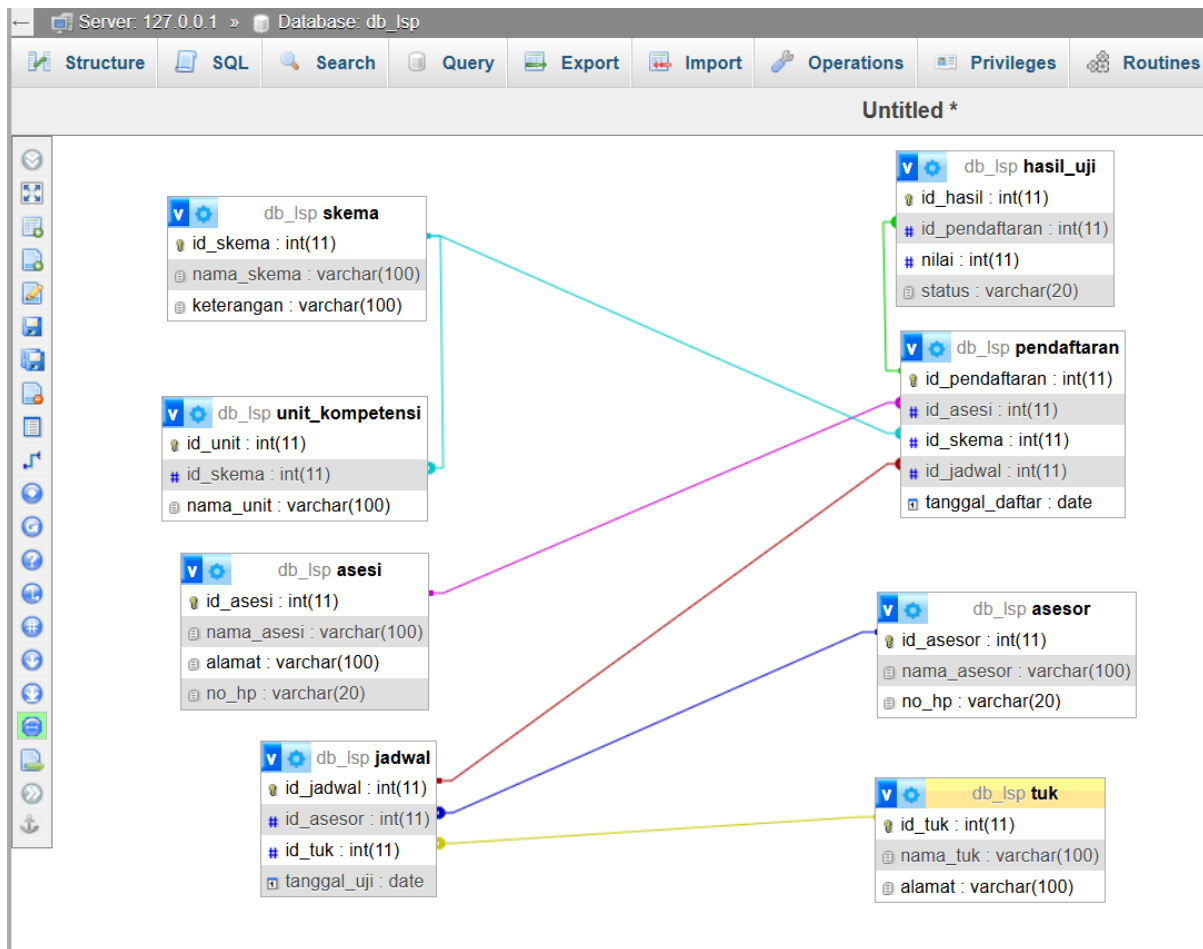


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TI 2A

POLITEKNIK PURBAYA TEGAL
TEKNIK INFORMATIKA 2025/2026

1. ERD (Entity Relationship Diagram)



2. Implementasi Database

```
MariaDB [(none)]> CREATE DATABASE db_lsp;  
Query OK, 1 row affected (0.003 sec)  
  
MariaDB [(none)]> USE db_lsp;  
Database changed  
MariaDB [db_lsp]> _
```

3. Membuat Seluruh Tabel

Tabel asesor

```
MariaDB [db_lsp]> CREATE TABLE asesor (  
  -> id_asesor INT AUTO_INCREMENT PRIMARY KEY,  
  -> nama_asesor VARCHAR(100),  
  -> no_hp VARCHAR(20)  
  -> );  
Query OK, 0 rows affected (0.020 sec)
```

Tabel asesi

```
MariaDB [db_lsp]> CREATE TABLE asesi (  
  -> id_asesi INT AUTO_INCREMENT PRIMARY KEY,  
  -> nama_asesi VARCHAR(100),  
  -> alamat VARCHAR(100),  
  -> no_hp VARCHAR(20)  
  -> );  
Query OK, 0 rows affected (0.015 sec)
```

Tabel skema

```
MariaDB [db_lsp]> CREATE TABLE skema (  
  -> id_skema INT AUTO_INCREMENT PRIMARY KEY,  
  -> nama_skema VARCHAR(100),  
  -> keterangan VARCHAR(100)  
  -> );  
Query OK, 0 rows affected (0.011 sec)
```

Tabel unit kompetensi

```
MariaDB [db_lsp]> CREATE TABLE unit_kompetensi (  
  -> id_unit INT AUTO_INCREMENT PRIMARY KEY,  
  -> id_skema INT,  
  -> nama_unit VARCHAR(100),  
  -> FOREIGN KEY (id_skema)  
  -> REFERENCES skema(id_skema)  
  -> );  
Query OK, 0 rows affected (0.047 sec)
```

Tabel tuk

```
MariaDB [db_lsp]> CREATE TABLE tuk (  
  -> id_tuk INT AUTO_INCREMENT PRIMARY KEY,  
  -> nama_tuk VARCHAR(100),  
  -> alamat VARCHAR(100)  
  -> );  
Query OK, 0 rows affected (0.017 sec)
```

Tabel jadwal

```
MariaDB [db_lsp]> CREATE TABLE jadwal (  
  -> id_jadwal INT AUTO_INCREMENT PRIMARY KEY,  
  -> id_asesor INT,  
  -> id_tuk INT,  
  -> tanggal_uji DATE,  
  -> FOREIGN KEY (id_asesor)  
  -> REFERENCES asesor(id_asesor),  
  -> FOREIGN KEY (id_tuk)  
  -> REFERENCES tuk(id_tuk)  
  -> );  
Query OK, 0 rows affected (0.046 sec)
```

Tabel pendaftaran

```
MariaDB [db_lsp]> CREATE TABLE pendaftaran (  
  -> id_pendaftaran INT AUTO_INCREMENT PRIMARY KEY,  
  -> id_asesi INT,  
  -> id_skema INT,  
  -> id_jadwal INT,  
  -> tanggal_daftar DATE,  
  -> FOREIGN KEY (id_asesi)  
  -> REFERENCES asesesi(id_asesi),  
  -> FOREIGN KEY (id_skema)  
  -> REFERENCES skema(id_skema),  
  -> FOREIGN KEY (id_jadwal)  
  -> REFERENCES jadwal(id_jadwal)  
  -> );  
Query OK, 0 rows affected (0.050 sec)
```

Tabel hasil uji

```
MariaDB [db_lsp]> CREATE TABLE hasil_uji (  
  -> id_hasil INT AUTO_INCREMENT PRIMARY KEY,  
  -> id_pendaftaran INT,  
  -> nilai INT,  
  -> status VARCHAR(20),  
  -> FOREIGN KEY (id_pendaftaran)  
  -> REFERENCES pendaftaran(id_pendaftaran)  
  -> );  
Query OK, 0 rows affected (0.066 sec)
```

4. Masukan Data

```
MariaDB [db_lsp]> INSERT INTO asesor VALUES  
  -> (NULL, 'Ahmad Fauzi', '0811111111'),  
  -> (NULL, 'Budi Santoso', '0811111112'),  
  -> (NULL, 'Citra Dewi', '0811111113'),  
  -> (NULL, 'Deni Saputra', '0811111114'),  
  -> (NULL, 'Eka Prasetyo', '0811111115');  
Query OK, 5 rows affected (0.010 sec)  
Records: 5 Duplicates: 0 Warnings: 0
```

5 asesor

```

MariaDB [db_lsp]> INSERT INTO asesi VALUES
-> (NULL,'Andi','Tegal','081'),
-> (NULL,'Bambang','Brebes','082'),
-> (NULL,'Candra','Pemalang','083'),
-> (NULL,'Dian','Slawi','084'),
-> (NULL,'Eko','Tegal','085'),
-> (NULL,'Fajar','Brebes','086'),
-> (NULL,'Galih','Slawi','087'),
-> (NULL,'Hendra','Pemalang','088'),
-> (NULL,'Indra','Tegal','089'),
-> (NULL,'Joko','Brebes','090'),
-> (NULL,'Kiki','Slawi','091'),
-> (NULL,'Lina','Pemalang','092'),
-> (NULL,'Maya','Tegal','093'),
-> (NULL,'Nina','Brebes','094'),
-> (NULL,'Oki','Slawi','095'),
-> (NULL,'Putra','Pemalang','096'),
-> (NULL,'Qori','Tegal','097'),
-> (NULL,'Rina','Brebes','098'),
-> (NULL,'Santi','Slawi','099'),
-> (NULL,'Tono','Pemalang','100');
Query OK, 20 rows affected (0.017 sec)
Records: 20  Duplicates: 0  Warnings: 0

```

20 asesi

```

MariaDB [db_lsp]> INSERT INTO skema VALUES
-> (NULL,'Junior Programmer','IT'),
-> (NULL,'Web Developer','IT'),
-> (NULL,'Database Administrator','IT'),
-> (NULL,'Network Administrator','IT'),
-> (NULL,'Digital Marketing','Bisnis');
Query OK, 5 rows affected (0.010 sec)
Records: 5  Duplicates: 0  Warnings: 0

```

5 skema

```

MariaDB [db_lsp]> INSERT INTO unit_kompetensi VALUES
-> (NULL,1,'Algoritma'),
-> (NULL,1,'Pemrograman Dasar'),
-> (NULL,1,'OOP'),
-> (NULL,1,'Debugging'),
-> (NULL,2,'HTML'),
-> (NULL,2,'CSS'),
-> (NULL,2,'JavaScript'),
-> (NULL,2,'PHP'),
-> (NULL,3,'SQL Dasar'),
-> (NULL,3,'SQL Lanjut'),
-> (NULL,3,'Backup'),
-> (NULL,3,'Recovery'),
-> (NULL,4,'TCP/IP'),
-> (NULL,4,'Routing'),
-> (NULL,4,'Switching'),
-> (NULL,4,'Keamanan Jaringan'),
-> (NULL,5,'SEO'),
-> (NULL,5,'SEM'),
-> (NULL,5,'Content Marketing'),
-> (NULL,5,'Social Media');
Query OK, 20 rows affected (0.013 sec)
Records: 20  Duplicates: 0  Warnings: 0

```

20 unit kompetensi

```

MariaDB [db_lsp]> INSERT INTO tuk VALUES
  -> (NULL,'TUK Purbaya','Tegal'),
  -> (NULL,'TUK Brebes','Brebes'),
  -> (NULL,'TUK Slawi','Slawi'),
  -> (NULL,'TUK Pemalang','Pemalang'),
  -> (NULL,'TUK Pekalongan','Pekalongan');
Query OK, 5 rows affected (0.010 sec)
Records: 5 Duplicates: 0 Warnings: 0

```

5 tuk

```

MariaDB [db_lsp]> INSERT INTO jadwal VALUES
  -> (NULL,1,1,'2026-07-25'),
  -> (NULL,2,2,'2026-07-26'),
  -> (NULL,3,3,'2026-07-27'),
  -> (NULL,4,4,'2026-07-28'),
  -> (NULL,5,5,'2026-07-29'),
  -> (NULL,1,2,'2026-08-01'),
  -> (NULL,2,3,'2026-08-02'),
  -> (NULL,3,4,'2026-08-03'),
  -> (NULL,4,5,'2026-08-04'),
  -> (NULL,5,1,'2026-08-05');
Query OK, 10 rows affected (0.013 sec)
Records: 10 Duplicates: 0 Warnings: 0

```

10 jadwal

```

MariaDB [db_lsp]> INSERT INTO pendaftaran
  -> (id_asesi,id_skema,id_jadwal,tanggal_daftar)
  -> VALUES
  -> (1,1,1,'2026-07-20'),
  -> (2,2,2,'2026-07-20'),
  -> (3,3,3,'2026-07-20'),
  -> (4,4,4,'2026-07-20'),
  -> (5,5,5,'2026-07-20'),
  -> (6,1,6,'2026-07-20'),
  -> (7,2,7,'2026-07-20'),
  -> (8,3,8,'2026-07-20'),
  -> (9,4,9,'2026-07-20'),
  -> (10,5,10,'2026-07-20'),
  -> (11,1,1,'2026-07-20'),
  -> (12,2,2,'2026-07-20'),
  -> (13,3,3,'2026-07-20'),
  -> (14,4,4,'2026-07-20'),
  -> (15,5,5,'2026-07-20'),
  -> (16,1,6,'2026-07-20'),
  -> (17,2,7,'2026-07-20'),
  -> (18,3,8,'2026-07-20'),
  -> (19,4,9,'2026-07-20'),
  -> (20,5,10,'2026-07-20');
Query OK, 20 rows affected (0.014 sec)
Records: 20 Duplicates: 0 Warnings: 0

```

20 pendaftar

```

MariaDB [db_lsp]> INSERT INTO hasil_uji
  -> (id_pendaftaran,nilai,status)
  -> VALUES
  -> (1,85,'Kompeten'),
  -> (2,90,'Kompeten'),
  -> (3,75,'Kompeten'),
  -> (4,60,'Belum Kompeten'),
  -> (5,95,'Kompeten'),
  -> (6,88,'Kompeten'),
  -> (7,70,'Kompeten'),
  -> (8,65,'Belum Kompeten'),
  -> (9,92,'Kompeten'),
  -> (10,78,'Kompeten'),
  -> (11,84,'Kompeten'),
  -> (12,89,'Kompeten'),
  -> (13,73,'Kompeten'),
  -> (14,67,'Belum Kompeten'),
  -> (15,96,'Kompeten'),
  -> (16,82,'Kompeten'),
  -> (17,71,'Kompeten'),
  -> (18,69,'Belum Kompeten'),
  -> (19,91,'Kompeten'),
  -> (20,80,'Kompeten');
Query OK, 20 rows affected (0.012 sec)
Records: 20  Duplicates: 0  Warnings: 0

```

20 hasil uji

5. Buat Sebuah VIEW

```

MariaDB [db_lsp]> CREATE VIEW vw_peserta_sertifikasi AS
  -> SELECT
  -> a.nama_asesi,
  -> s.nama_skema,
  -> asr.nama_asesor,
  -> t.nama_tuk,
  -> h.nilai
  -> FROM pendaftaran p
  -> JOIN asesi a ON p.id_asesi=a.id_asesi
  -> JOIN skema s ON p.id_skema=s.id_skema
  -> JOIN jadwal j ON p.id_jadwal=j.id_jadwal
  -> JOIN asesor asr ON j.id_asesor=asr.id_asesor
  -> JOIN tuk t ON j.id_tuk=t.id_tuk
  -> JOIN hasil_uji h ON p.id_pendaftaran=h.id_pendaftaran;
Query OK, 0 rows affected (0.014 sec)

```

Menampilkan VIEW

```
MariaDB [db_lsp]> SELECT * FROM vw_peserta_sertifikasi;
```

nama_asesi	nama_skema	nama_asesor	nama_tuk	nilai
Andi	Junior Programmer	Ahmad Fauzi	TUK Purbaya	85
Fajar	Junior Programmer	Ahmad Fauzi	TUK Brebes	88
Kiki	Junior Programmer	Ahmad Fauzi	TUK Purbaya	84
Putra	Junior Programmer	Ahmad Fauzi	TUK Brebes	82
Bambang	Web Developer	Budi Santoso	TUK Brebes	90
Galih	Web Developer	Budi Santoso	TUK Slawi	70
Lina	Web Developer	Budi Santoso	TUK Brebes	89
Qori	Web Developer	Budi Santoso	TUK Slawi	71
Candra	Database Administrator	Citra Dewi	TUK Slawi	75
Hendra	Database Administrator	Citra Dewi	TUK Pemalang	65
Maya	Database Administrator	Citra Dewi	TUK Slawi	73
Rina	Database Administrator	Citra Dewi	TUK Pemalang	69
Dian	Network Administrator	Deni Saputra	TUK Pemalang	60
Indra	Network Administrator	Deni Saputra	TUK Pekalongan	92
Nina	Network Administrator	Deni Saputra	TUK Pemalang	67
Santi	Network Administrator	Deni Saputra	TUK Pekalongan	91
Eko	Digital Marketing	Eka Prasetyo	TUK Pekalongan	95
Joko	Digital Marketing	Eka Prasetyo	TUK Purbaya	78
Oki	Digital Marketing	Eka Prasetyo	TUK Pekalongan	96
Tono	Digital Marketing	Eka Prasetyo	TUK Purbaya	80

```
20 rows in set (0.002 sec)
```

6. Gunakan Subquery

A. Menampilkan peserta dengan nilai di atas rata-rata

```
MariaDB [db_lsp]> SELECT a.nama_asesi,h.nilai
-> FROM asesi a
-> JOIN pendaftaran p
-> ON a.id_asesi=p.id_asesi
-> JOIN hasil_uji h
-> ON p.id_pendaftaran=h.id_pendaftaran
-> WHERE h.nilai >
-> (
-> SELECT AVG(nilai)
-> FROM hasil_uji
-> );
```

nama_asesi	nilai
Andi	85
Bambang	90
Eko	95
Fajar	88
Indra	92
Kiki	84
Lina	89
Oki	96
Putra	82
Santi	91

```
10 rows in set (0.002 sec)
```


B. Menampilkan asesor yang belum memiliki jadwal

```
MariaDB [db_lsp]> SELECT *  
-> FROM asesor  
-> WHERE id_asesor NOT IN  
-> (  
-> SELECT id_asesor  
-> FROM jadwal  
-> );  
Empty set (0.008 sec)
```

C. Menampilkan skema dengan peserta terbanyak

```
MariaDB [db_lsp]> SELECT nama_skema  
-> FROM skema  
-> WHERE id_skema =  
-> (  
-> SELECT id_skema  
-> FROM pendaftaran  
-> GROUP BY id_skema  
-> ORDER BY COUNT(*) DESC  
-> LIMIT 1  
-> );  
+-----+  
| nama_skema |  
+-----+  
| Junior Programmer |  
+-----+  
1 row in set (0.001 sec)
```